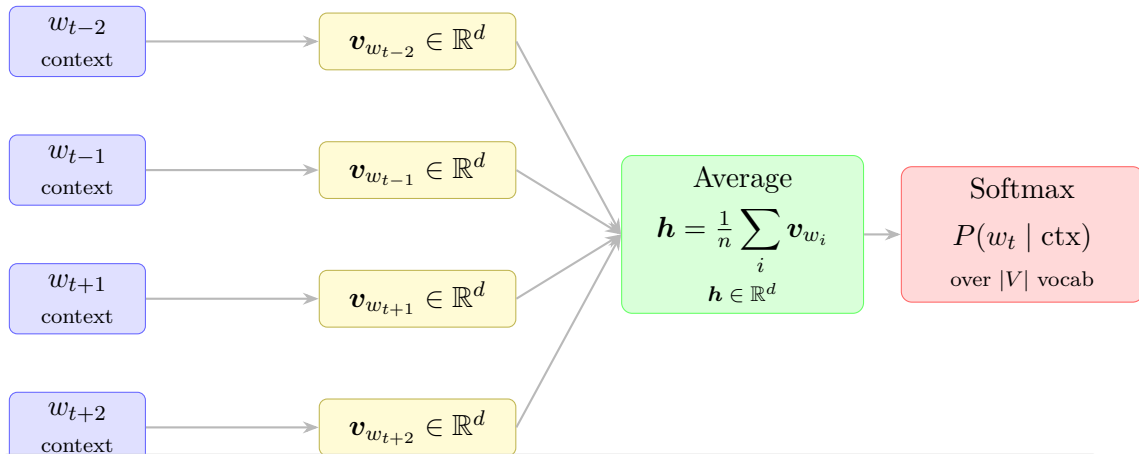


Embedding lookup ($\mathbf{W}_{\text{in}}$)



CBOW Objective:
$$P(w_t \mid \text{context}) = \frac{\exp(\mathbf{u}_{w_t}^\top \cdot \mathbf{h})}{\sum_{k \in V} \exp(\mathbf{u}_k^\top \cdot \mathbf{h})}$$

$\mathbf{u}_w$ = output embedding, $\mathbf{v}_w$ = input embedding (two separate weight matrices $\mathbf{W}_{\text{in}}, \mathbf{W}_{\text{out}} \in \mathbb{R}^{|V| \times d}$)